Rui Tian

Mail: ruitian12@gmail.com

Education

- **Fudan University** — Shanghai, China
  Ph.D., College of Computer Science and Artificial Intelligence. — *Sep. 2021 ~ Jun. 2026*
  Supervisors: Dr. Zuxuan Wu and Dr. Yu-Gang Jiang.
- **Fudan University** — Shanghai, China
  Bachelor of Science (Elite Program), School of Computer Science. — *Sep. 2017 ~ Jun. 2021*

Research Experience

- **Research Intern at Apple.**
  Supervisor: Dr. Mingfei Gao. Multimodal LLMs. — *Jan. 2026 ~ Jun. 2026*
- **Research Intern at Apple.**
  Supervisors: Dr. Mingfei Gao and Dr. Mingze Xu. Multimodal LLMs. — *Nov. 2024 ~ Nov. 2025*
- **Research Intern at Microsoft Research Asia.**
  Supervisor: Dr. Qi Dai. Generative AI. — *Apr. 2023 ~ Oct. 2024*
- **Research Intern at Microsoft Research Asia.**
  Supervisor: Dr. Qi Dai. Scalable Vision. — *Oct. 2021 ~ Sep. 2022*

Publications & Preprints

* denotes equal contribution.

- **[Preprint 2026] MintAct: A Unified Visual Agent for Digital Environments.** arXiv
  Mingfei Gao*, **Rui Tian***, Haiming Gang*, Bohan Zhai*, Le Zhang, Yuanzheng Gong, Di Feng, Ege Özsoy, Kaixin Ma, Vishwesh Kirthivasan, Oğuzhan Fatih Kar, Roman Bachmann, Anders Boesen Lindbo Larsen, Afshin Dehghan.
  We present MintAct, a family of vision-language models that unifies UI grounding, multi-step navigation, and visual tool use across mobile, desktop, and web environments. We develop scalable environments and asynchronous reinforcement learning to train generalist agents that match domain specialists and achieve 48.9 on OSWorld-Verified.
- **[CVPR 2026] UniGen-1.5: Enhancing Image Generation and Editing through Reward Unification in Reinforcement Learning.** arXiv
  Rui Tian, Mingfei Gao, Haiming Gang, Jiasen Lu, Zhe Gan, Yinfei Yang, Zuxuan Wu, Afshin Dehghan.
  We present UniGen-1.5 for unified image understanding, generation, and editing. A shared-reward reinforcement learning strategy jointly improves generation and editing, while a lightweight Edit Instruction Alignment stage strengthens instruction comprehension.
- **[NeurIPS 2025] UniGen: Enhanced Training & Test-Time Strategies for Unified Multimodal Understanding and Generation.** arXiv, GitHub
  Rui Tian*, Mingfei Gao*, Mingze Xu*, Jiaming Hu, Jiasen Lu, Zuxuan Wu, Yinfei Yang, Afshin Dehghan.
  We propose UniGen, a unified multimodal large language model for both understanding and generation, and design a new Chain-of-Thought Verification (CoT-V) strategy by enabling it to act as both an image generator and a step-by-step image quality verifier.

- **[ICCV 2025] REDUCIO! Generating 1024×1024 Video within 16 Seconds using Extremely Compressed Motion Latents. arXiv, GitHub**

  Rui Tian, Qi Dai, Jianmin Bao, Kai Qiu, Yifan Yang, Chong Luo, Zuxuan Wu, Yu-Gang Jiang. We propose Reducio, a framework built end-to-end for efficient high-resolution video generation based on a VAE with an extreme spatiotemporal down-sampling ratio of $32 \times 32 \times 4$.

- **[NeurIPS 2024] DeepStack: Deeply Stacking Visual Tokens is Surprisingly Simple and Effective for LMMs. arXiv, GitHub**

  Lingchen Meng, Jianwei Yang, **Rui Tian**, Xiyang Dai, Zuxuan Wu, Jianfeng Gao, Yu-Gang Jiang.

  We introduce DeepStack, a simple strategy that connects vision and language transformers by injecting visual tokens into multiple LLM layers, improving multimodal understanding.

- **[CVPR 2023] ResFormer: Scaling ViTs with Multi-Resolution Training. arXiv, GitHub**

  Rui Tian, Zuxuan Wu, Qi Dai, Han Hu, Yu Qiao, Yu-Gang Jiang.

  We introduce ResFormer to improve vision transformer performance across image resolutions, including unseen resolutions, through multi-resolution training, scale consistency loss, and adaptive global-local positional embeddings.

- **[TMM 2025] The Role of ViT Design and Training in Robustness to Common Corruptions. IEEE**

  Rui Tian, Zuxuan Wu, Qi Dai, Micah Goldblum, Han Hu, Yu-Gang Jiang.

  We evaluate the robustness of vision transformers to common corruptions and identify architectural choices that improve robustness. We further introduce input-conditioned dynamic data augmentation.

- **[TPAMI 2026] TinyMem: Condensing Multimodal Memory for Long-Form Video Action Detection. IEEE**

  Rui Tian, Qi Dai, Han Hu, Zuxuan Wu.

  We introduce TinyMem for long-form video action detection, combining succinct textual descriptions with region-condensed visual memory to capture long-range dependencies efficiently.

Academic Service

- Distinguished Reviewer for CVPR 2025.
- Reviewer for ICCV 2023, ICLR 2024 - 2025, ICML 2024, CVPR 2023 - 2026, NeurIPS 2023 - 2025, AAAI 2026, TMLR.

Honors and Awards

- *Outstanding Graduate*, Fudan University, 2026.
- *First-Class PhD Academic Scholarship*, Fudan University, 2024 - 2025.
- *Junzheng Scholarship*, Undergraduate Research Opportunities Program (UROP), 2021.